

Autonomous Diffusion: locked-test verification on CIFAR-10

10,000 samples / cell, 5 seeds, 17 samplers, 510 runs

thermau5

May 28, 2026

Status. Locked test complete. The contract (outputs/freeze_record.yaml, sha256 450925ba0791...) was frozen on 2026-05-24 and not modified during the 45.77 h evaluation. The critic guards refused any drift; 510/510 runs succeeded, zero failures. Per-cell numbers below are mean $\pm$ SEM over 5 declared seeds $\{0, 1, 2, 3, 4\}$. No tuning post-freeze.

1 Theory

Setup. Let $p_0 = \mathcal{N}(0, I)$ be the prior and $p_1 = p_{\text{data}}$ the data distribution. The generative ODE in noise coordinate σ is

$$\frac{dx}{d\sigma} = g(x, \sigma; \theta), \quad g(x, \sigma; \theta) = \frac{x - D_\theta(x, \sigma)}{\sigma},$$

with D_θ the pretrained EDM denoiser. A *sampler* is the discretized controlled flow

$$x_{i+1} = F_s(x_i, \sigma_i, \sigma_{i+1}; \theta, \mathcal{H}_i), \quad i = 0, \dots, K-1,$$

specified by the tuple

$$u = (\theta, \pi, s, m, N)$$

where θ is the backbone, π is the path/coupling family, s the solver core (update law F_s), $m(\sigma)$ the sampling-grid density inducing $\sigma_0 > \sigma_1 > \dots > \sigma_K = 0$, and N the cost budget (NFE / wall-clock).

Schedule equivalence under coordinate change. The names *noise schedule*, *time schedule*, and *grid schedule* refer to the same mathematical object m viewed in different coordinates. A noise schedule specifies $\sigma(t)$ along a continuous time axis; the inverse $t(\sigma)$ is the time schedule on the noise axis; and the discrete grid $\{\sigma_i\}_{i=0}^K$ is the quantile sampling of either, related to m by inverse-CDF $\sigma_i = M^{-1}(i/K)$ where $M(\sigma) = \int_\sigma^B m(s) ds$. Under any smooth monotone reparameterization $\varphi: \sigma \mapsto \sigma'$, the certificate's two integrals transform covariantly — m , d_s , and a all transform as densities — and the closed-form optimizer $m_s^*(\sigma) \propto d_s(\sigma)^{1/(p+1)}$ is invariant up to the same change of variable. So Karras's ρ -parameterized noise schedule, the DDPM β -schedule on time, and our discrete grid $\{\sigma_i\}$ are equivalent control objects; the theory and the implementation can be stated in whichever coordinate is most convenient for a given solver. We use σ -density throughout because the EDM net is naturally parameterized in σ .

Certified KL/NLL risk. The certificate decomposes the terminal-risk bound additively:

$$R(u) \leq Q_s(u) = Q_0 + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_s(r)}{m(r)^p} dr,$$

where ρ is the training/path-time allocation (frozen by the pretrained net for this work), $a(r)$ is the per- r statistical weight, $d_s(r)$ is the solver-specific discretization density at noise level r , and p is the solver order (Heun: $p = 2$).

Derivation of ρ^* (Cauchy–Schwarz). The statistical sub-problem is $\min_{\rho} \int_0^B a(r)/\rho(r) dr$ subject to $\int_0^B \rho(r) dr = 1$. By Cauchy–Schwarz,

$$\left(\int_0^B \sqrt{a(r)} dr \right)^2 = \left(\int_0^B \sqrt{\frac{a(r)}{\rho(r)}} \sqrt{\rho(r)} dr \right)^2 \leq \left(\int_0^B \frac{a(r)}{\rho(r)} dr \right) \left(\int_0^B \rho(r) dr \right),$$

so $\int a/\rho dr \geq (\int \sqrt{a} dr)^2$ with equality iff

$$\rho^*(r) = \frac{\sqrt{a(r)}}{\int_0^B \sqrt{a(s)} ds}.$$

Equivalently the optimal noise schedule satisfies $\beta^*(t) = \frac{dr}{dt} = \frac{\int_0^B \sqrt{a(s)} ds}{\sqrt{a(r(t))}}$, which strictly subsumes hand-crafted families: constant a gives uniform noise-time, a peaking near $\log\text{-SNR}=0$ recovers $\log\text{-SNR}$ -focused schedules, and endpoint-difficulty a adapts off the cosine/Karras fixed curves. In this work ρ is frozen by the pretrained EDM net, but the bound is exact when the training distribution matches ρ^* .

Derivation of m_s^* (Lagrangian KKT). The discretization sub-problem is $\min_m \int_0^B d_s(r)/m(r)^p dr$ subject to $\int_0^B m(r) dr = N$. The Lagrangian $\mathcal{L} = \int d_s/m^p dr - \lambda(\int m dr - N)$ has stationary point $-p d_s(r)/m(r)^{p+1} - \lambda = 0$, giving

$$m_s^*(r) = \frac{N d_s(r)^{1/(p+1)}}{\int_0^B d_s(s)^{1/(p+1)} ds}, \quad \Delta r^*(r) \propto d_s(r)^{-1/(p+1)}.$$

Hence the closed-form Pareto-optimal sampling-grid density is

$$\boxed{m_s^*(\sigma) \propto d_s(\sigma)^{1/(p+1)}}$$

For each solver core $s \in \{\text{Heun}, \text{DPM-Solver++}, \text{UniPC}, \text{DEIS}, \text{Restart}\}$ we run `proposed_<core>` with its actual update law and the certificate-optimal grid m_s^* above. The implementation also applies a Karras-style perceptual weight $w(\sigma) = \sigma^{-k}$ with $k = 2$ (the only validation-tuned hyperparameter) so the effective form is $m_s^*(\sigma) \propto (d_s(\sigma) \cdot w(\sigma))^{1/(p+1)}$. For Euler ($p = 1$) the optimal local step satisfies $\Delta r^*(r) \propto d_s(r)^{-1/2}$.

Minimized certificate and minimum NFE. Substituting ρ^* and m_s^* into Q_s gives the minimized certificate

$$Q_s^*(N) = Q_0 + \frac{c_{\text{stat}}}{n} \left(\int_0^B \sqrt{a(r)} dr \right)^2 + c_{\text{disc}} \frac{\left(\int_0^B d_s(r)^{1/(p+1)} dr \right)^{p+1}}{N^p}.$$

Inverting for a target accuracy ε yields the accuracy-constrained minimum NFE

$$N^*(\varepsilon) = \left[\frac{c_{\text{disc}} \left(\int_0^B d_s(r)^{1/(p+1)} dr \right)^{p+1}}{\varepsilon - Q_0 - \frac{c_{\text{stat}}}{n} \left(\int_0^B \sqrt{a(r)} dr \right)^2} \right]^{1/p}.$$

Pareto-domination theorem. For any baseline schedule/grid $u_b = (\rho_b, m_b)$ with NFE N_b , Cauchy–Schwarz gives $\int a/\rho_b dr \geq (\int \sqrt{a} dr)^2$, and the KKT step gives $\int d_s/m_b^p dr \geq (\int d_s^{1/(p+1)} dr)^{p+1}/N_b^p$, so $Q_s(u_b) \geq Q_s^*(N_b)$. Setting $\varepsilon = Q_s(u_b)$ makes

$$N^*(\varepsilon) \leq N_b, \quad Q_s(u^*) \leq Q_s(u_b),$$

i.e. the certificate-optimal scheduler weakly Pareto-dominates the baseline under the certified accuracy–efficiency criterion. The inequality is strict unless the baseline itself is the optimizer of the same certificate.

Solver-specific discretization density. $d_s(\sigma)$ is the per- σ local-truncation-error density of the chosen solver s . Empirically (full recipe in Appendix A): take a batch of in-distribution noisy images at σ_a , step them once with the actual update law F_s to σ_b , compare against a 16-substep Heun reference, and normalize by interval length:

$$d_s(\sigma_i) = \frac{1}{|\sigma_a - \sigma_b|} \frac{1}{B} \sum_{b=1}^B \|F_s(x_b; \sigma_a \rightarrow \sigma_b) - \text{Heun}^{(16)}(x_b; \sigma_a \rightarrow \sigma_b)\|_{\text{RMS}}$$

on a uniform-log- σ grid of $M = 32$ intervals over $[\sigma_{\min}, \sigma_{\max}] = [0.002, 80]$. The reference is solver-independent; the test step is the actual F_s , so d_s is solver-specific by construction.

Bootstrap: where x_b comes from. The calibration needs in-distribution noisy images x_b at each σ_a *before* the certificate has been solved. We bootstrap by running the standard (Heun, Karras $\rho=7$) sampler — the hand-tuned baseline from Karras 2022 that already produces $\text{FID} \approx 2$ at 35 NFE — from pure noise at $\sigma_{\max}$ down to $\sigma_{\min}$, and use its trajectory x_b as the calibration source. This is the *reverse* (sampler) trajectory, not the forward noising process: the EDM-Heun-Karras run is an off-the-shelf reference that doesn’t depend on m_s^* , so the dependency chain is acyclic. Bootstrap cost is one-time per pretrained net (~ 10 k NFE, ~ 30 s on the 4090, cached). Iterating the bootstrap with the new (s, m_s^*) sampler is possible but empirically unnecessary at validation-grade NFE.

Scope (Appendix A correction). The boxed pointwise form applies to single-step / pointwise-defect solvers (RK-class: Heun, EDM-Euler, DPM-Solver-2) and to multistep solvers whose leading defect is empirically pointwise (DPM-Solver++ 2M, UniPC — Theorem A). For q -step Adams–Bashforth solvers (DEIS tAB-2, PNDM/PLMS-4) the local residual is the history-coupled interpolation-kernel error $e_i^{\text{AB}} \propto g^{(q)}(\xi) \prod_{\ell=0}^{q-1} (\sigma - \sigma_{i-\ell})$, which does not factor into a scalar $d_s(\sigma)$. The certificate is then replaced by a coupled-sequence functional $D_s^{\text{AB}}(\Pi_N)$ and the optimal grid is found by sequence optimization $\Pi_{N,s}^* = \arg \min_{\Pi_N} D_s^{\text{AB}}(\Pi_N)$ (Theorem B, implemented in this work).

Headline empirical result. On the locked CIFAR-10 test split (10k samples / cell, 5 seeds, 510 runs, contract sha256 450925ba0791...):

$$(s_*, m_{s_*}^*) = (\text{UniPC}, \text{Ours}) \quad \text{owns the entire cost-honest Pareto frontier at} \quad \text{actual NFE} \in [5, 32]$$

with 6 of 9 frontier points held by (Ours, s) variants and the largest single-cell improvement -31.7 FID at Heun NFE = 9.

2 Comparison Hierarchy

A generative sampler is the tuple

$$u = (\theta, \pi, s, m, N)$$

where θ is the backbone (architecture + weights), π is the path/coupling family (which generative ODE/SDE the backbone parameterizes), s is the solver core (the discrete update law F_s), m is the schedule/grid (where on the noise axis F_s is evaluated), and N is the cost budget (actual NFE or wall-clock). Our method is not a new generative model; it is per-solver-core schedule control:

$$m_s^*(\sigma) \propto d_s(\sigma)^{1/(p+1)}.$$

We organize experiments by which variable changes. The table below uses the convention that throughout this report each experiment has a triple [FIX | VARIABLE | SELECT]: FIX are background objects held constant, VARIABLE is the swept axis, SELECT is what the experiment chooses as its outcome (winning member, best-per-NFE, etc.).

Level	Variable	Fixed objects	Question	Status
1	s	$\theta, \pi, m_s^{default}$	Strongest solver baseline?	done
2	m	θ, π, s	Does m_s^* beat $m_s^{default}$?	done (heuristic, tuned, learned)
3	π	comparable backbone/compute	Does path-straightening beat schedule control?	<i>open</i> (queued, see §5)
4	θ	none / broad protocol	Full generative-model leaderboard	<i>out of scope</i>

Baselines included at each Level (each row is a distinct (s, m, π) configuration on the same θ_{EDM} backbone except where noted):

Level	axis tested	baselines in the locked test (+ post-freeze additions)
1	s (solver core) at fixed Karras- $\rho=7$ grid	<code>edm_euler</code> , <code>edm_heun</code> , <code>ddim</code> , <code>ddpm_ancestral</code> , <code>dpm_solver</code> , <code>dpm_solver_pp</code> , <code>unipc</code> , <code>deis</code> , <code>pndm</code> , <code>restart</code> (10 cores in locked test)
2a (fixed-heuristic m)	m class $\in \{\text{Karras } \rho=7, \text{uniform-log}\}$	<code>karras_schedule</code> , <code>uniform_schedule</code> (the two heuristic m classes paired with $s=\text{Heun}$)
2b (tuned-parametric m)	$m = \text{Karras-}\rho^*$, ρ chosen on validation	<code>rho_star_unipc</code> (UniPC core; Issue #2 §3.1; §6). <i>Open on DPM Solver++, DEIS, Heun cores.</i>
2c (learned m)	$m = \text{AYS}$ (Gaussian-closed-form KLUB, paper-faithful)	<code>ays_unipc</code> (Round 5b, post-freeze addition; §4). <i>Open on other cores.</i>
2d (nonparametric-calibrated m , ours)	$m = m_s^* \propto d_s^{1/(p+1)}$	<code>proposed_heun</code> , <code>proposed_dpmp</code> , <code>proposed_unipc</code> , <code>proposed_deis</code> , <code>proposed_restart</code> (5 cores in locked test) <i>none in locked test</i> . Queued for Round 6: Rectified Flow, Re flow, OT-CFM (pretrained ckpts available); MSFM, OFM (training needed). See §5 for the plan.
3	π (path/coupling): backbone retrained for non-EDM path	

The locked test in this report covers Level 1 + Level 2a + Level 2d under $\theta = \theta_{EDM}$, $\pi = \pi_{EDM}$. Levels 2b (tuned-parametric) and 2c (learned-schedule) are partial: implemented for $s=\text{UniPC}$, open on other cores. Level 3 is not in the locked test and is the explicit next round before any path-straightening claim or scaling to FFHQ / AFHQv2 / Wasserstein variants.

3 Baseline audit

Setup: FIX $\theta = \theta_{EDM}$, $\pi = \pi_{EDM}$; VARIABLE = the columns (s, m) of the table below; SELECT = none (an audit, not a contest). All rows are Level 1+2a+2d under the locked contract; rows `rho_star_unipc` and `ays_unipc` are post-freeze additions covering Levels 2b and 2c respectively (rows shown but shaded in their own subsections).

For each sampler used in the locked test, the per-experiment fixed/variable objects:

sampler id	path π	backbone θ	solver core s	baseline grid $m_s^{default}$	paired (Ours, s)	actual NFE
<code>edm_euler</code>	EDM diffusion	pretr. EDM	Euler	Karras $\rho=7$	–	K
<code>edm_heun</code>	EDM diffusion	pretr. EDM	Heun	Karras $\rho=7$	<code>proposed_heun</code>	$2K-1$
<code>karras_schedule</code>	EDM diffusion	pretr. EDM	Heun	Karras $\rho=7$	<code>proposed_heun</code>	$2K-1$
<code>uniform_schedule</code>	EDM diffusion	pretr. EDM	Heun	uniform-log	<code>proposed_heun</code>	$2K-1$
<code>ddim</code>	EDM diffusion	pretr. EDM	DDIM (1st)	DDPM β -derived	–	K
<code>ddpm_ancestral</code>	EDM diffusion	pretr. EDM	DDPM ancestral	DDPM β -derived	–	K
<code>dpm_solver</code>	EDM diffusion	pretr. EDM	DPM-Solver-2	Karras $\rho=7$	–	$2K-1$
<code>dpm_solver_pp</code>	EDM diffusion	pretr. EDM	DPM-Solver++ 2M	Karras $\rho=7$ in our impl.	<code>proposed_dpmp</code>	K
<code>unipc</code>	EDM diffusion	pretr. EDM	UniPC 2PC	Karras $\rho=7$ in our impl.	<code>proposed_unipc</code>	K
<code>deis</code>	EDM diffusion	pretr. EDM	DEIS tAB-2	Karras $\rho=7$ in our impl.	<code>proposed_deis</code>	K
<code>pndm</code>	EDM diffusion	pretr. EDM	PNDM/PLMS-4	Karras $\rho=7$ in our impl.	–	K +warmup
<code>restart</code>	EDM diffusion	pretr. EDM	Restart wrap. Heun	Xu et al. σ -band	<code>proposed_restart</code>	base + cycles

Wording note. The “baseline grid” column is the standard Karras / DDPM-derived grid used in *our EDM-compatible implementation*, not necessarily the exact grid each method’s original paper or release configures. The paired (Ours, s) variant runs the same solver core on the certificate-optimal grid m_s^* from this work. “–” means we have no paired variant (DDIM, DDPM, EDM-Euler, DPM-Solver-2, PNDM are baseline-only; the latter four are subsumed by their multistep / corrector variants on the proposed side).

4 Schedule (m) taxonomy

The grid $m(r)$ inducing $\sigma_0 > \sigma_1 > \dots > \sigma_K = 0$ takes one of five classes. Issue #2 in the repo demands this distinction explicitly:

class	example	description
fixed heuristic	Karras $\rho=7$, uniform-log- σ	fixed before validation/test
tuned parametric	Karras ρ^*	small parameter vector optimized on validation
learned scheduler	m_ϕ	trained schedule model/policy
adaptive scheduler	$m(x, r)$	sample/trajectory-dependent online schedule
nonparametric-calibrated	ours m_s^*	full calibrated profile $d_s(r)$ + closed-form certificate optimizer

Canonical statement (verbatim). Ours is a **nonparametric-calibrated scheduler**: it estimates a solver-specific difficulty profile $d_s(r)$ over the noise coordinate and then computes the certificate-optimal grid $m_s^*(r) \propto d_s(r)^{1/(p+1)}$. It is nonparametric *relative to low-dimensional schedule families* such as Karras ρ , because the calibrated object is a full profile $d_s(r)$, not a small parameter vector. It is calibrated *rather than learned* because $d_s(r)$ is measured from local solver error against a reference solver, not trained end-to-end by FID/test loss.

What the locked test already covers. fixed-heuristic and nonparametric-calibrated only:

- Karras $\rho=7$ (`karras_schedule`, `edm_heun`, `edm_euler`, `dpm_solver_pp`, `unipc`, `deis`, `pndm`, `restart` all run the Karras $\rho=7$ grid in our EDM-compatible implementation);
- Uniform-log- σ (`uniform_schedule`);
- DDPM- β -derived (`ddim`, `ddpm_ancestral`);
- Solver implementation defaults (`dpm_solver`, `restart`);
- Nonparametric-calibrated m_s^* (`proposed_heun`, `proposed_dpmp`, `proposed_unipc`, `proposed_deis`, `proposed_restart`).

What the locked test does not yet cover.

- **Tuned parametric.** No validation-optimized Karras ρ^* baseline was included. The locked test uses the fixed $\rho=7$ member of the Karras family for every Karras-grid sampler. *Round 5a (in progress) addresses this with a $\rho \in \{1, 2, 3, 5, 9, 12, 15\}$ sweep on UniPC.*
- **Learned scheduler.** No m_ϕ baseline (e.g. neural timestep policy).
- **Adaptive scheduler.** No sample-dependent $m(x, r)$ baseline.

Until a tuned parametric and a learned/adaptive schedule baseline are either implemented or marked infeasible with citation, the headline (s , m_s^*) win is to be interpreted as a win against *fixed-heuristic* schedules only, not against tuned/learned/adaptive ones.

Literature schedule-optimization baseline (Issue #2 §3.3). Closest published schedule-optimization methods and their fit for our fixed-EDM benchmark:

- **AYS (*Align Your Steps*)** (Sabour et al., ICML 2024, arXiv:2404.14507): optimizes a timestep grid by minimizing KL between the discrete sampler trajectory and a continuous-SDE reference. Released schedules: SDXL and SD 1.5 only; no CIFAR-10/EDM schedule released. **Reimplemented faithfully in this report (§4); appears in the headline AYS vs Karras vs (Ours, UniPC) table on the deterministic UniPC core.**
- **DDSS (*Watson et al. 2022, ICLR, arXiv:2202.05830*)**: differentiable sampler search by backpropagating through sample-quality scores. *Code-availability triage (2026-05-27)*: no public GitHub released; paper headlines LSUN-church, not CIFAR-10. *Status: no public code, mentioned for completeness.*
- **ACS / OC schedule** (Kong, Wang, Srikant 2026, arXiv:2605.21911, “Noise Schedule Design for Diffusion Models: An Optimal Control Perspective”; Issue #3 §1). Closed-form expressions for the optimal noise schedule under affine-coupled (f, g) parametrization, Corollary 1, under either strongly-log-concave or Gaussian-mixture data assumptions. Solver-agnostic. *Code-availability triage (2026-05-27)*: no public GitHub released; CIFAR-10 hyperparameters $(\theta, \omega, g_0, \gamma)$ not published. The schedule has a free scalar γ (parameterized via $c = \gamma\lambda_\star$ where $\lambda_\star = W(Kn)$ is a Lambert-W function of NFE budget). This places ACS at the *tuned-parametric* level of the m -taxonomy alongside Karras- $\rho^\star$ — not the learned or nonparametric-calibrated levels. *Status: no public code, mentioned for completeness.*
- **HSO** (Zhu et al. 2025, arXiv:2511.11688, “Hierarchical Schedule Optimization for Fast and Robust Diffusion Model Sampling”). Bi-level optimization with the Midpoint Error Proxy (MEP, solver-agnostic local objective) and Spacing-Penalized Fitness (SPF, global fitness with closeness penalty). *Code-availability triage (2026-05-27)*: no public GitHub released; paper headlines LAION-Aesthetics + SD 2.1, no CIFAR-10 result published. *Status: no public code, mentioned for completeness.*
- **ART / ART-RL** (Huang, Tang, Zhou 2026, arXiv:2601.18681, “ART for Diffusion Sampling: A Reinforcement Learning Approach to Timestep Schedule”). Adaptive Reparameterized Time + continuous-time RL with Gaussian policies; transfers from CIFAR-10 to FFHQ/AFHQv2/ ImageNet without retraining. “Within the official EDM pipeline” — likely Heun-class-bound. *Code-availability triage (2026-05-27)*: no public GitHub released; arXiv’d 2026-01-26 (very recent). *Status: no public code, mentioned for completeness.*
- **DPM-Solver-v3** (Zheng, Lu et al., NeurIPS 2023, arXiv:2310.13268). EMS-corrected ODE solver: three model-specific empirical statistics $(l_\lambda, s_\lambda, b_\lambda)$ are precomputed once from the pretrained model and inserted into the solver’s update law. The EMS lives *inside the solver*; the schedule is fully separable (the EMS solver can be run with any sigma grid). *Code-availability triage (2026-05-27)*: public code at github.com/thu-ml/DPM-Solver-v3. Paper reports unconditional CIFAR-10 FID 12.21 at NFE=5 and 2.51 at NFE=10 (protocol unverified vs our Clean-FID/10k). *Status: this is a Level-1 (new solver core) baseline, not Level-2; queued for Round 6 along with proposed_dpm_solver_v3 (our $m_s^\star$ on the same EMS-corrected core).*
- **GENIE** (Dockhorn et al. 2022). Distills a 2nd-order solver via training, learning solver weights and a schedule jointly at a specific NFE. The distilled solver expects its trained schedule; running it on a different schedule is not well-defined without redoing distillation. *Status: cite-only by genuine bundling (and additionally a different θ , so closer to Level-4).*

Triage summary. Of the seven published methods reviewed, only AYS has been reimplemented in this report (§4) because it is the only one with a paper-specified algorithm we could reproduce end-to-end from the published equations alone (Sec. 3, Lemma 3.3, hierarchical subdivision Sec. 3.3). DDSS, ACS, HSO, and ART have no public code as of the audit date (2026-05-27) and are mentioned for completeness only — not queued for reimplementation, because without released schedules or code the comparison would be against *our reinterpretation* of each method rather than against the method itself. **DPM-Solver-v3 is a Level-1 baseline (new solver core) with public code and is queued for Round 6;** its EMS lives inside the solver, not the schedule, so $(Ours, m_s^*)$ on the EMS-corrected core is the natural follow-on comparison. GENIE bundles solver+schedule with distilled weights (closer to Level-4) and is cite- only. The honest position is therefore: $(Ours, m_s^*)$ is the only nonparametric-calibrated schedule in the master Level-2 comparison; AYS (faithfully reproduced) is the only learned-schedule baseline evaluated here; DPM-Solver-v3 is the only further code-public Level-1 baseline pending Round 6; ACS / HSO / ART / DDSS / GENIE are cited but not benchmarked.

Solver compatibility of these methods. The four methods fall into three compatibility classes:

- *Drop-in (would work with our mainstream solvers).* AYS and DDSS output a fixed sigma vector $(\sigma_0, \dots, \sigma_K)$ that plugs into any solver core taking a sigma grid (Heun, DPM-Solver++, UniPC, DEIS). The only barrier is re-deriving the schedule on EDM CIFAR-10: AYS publishes only SDXL and SD 1.5 schedules; DDSS publishes CIFAR-10 fitted parameters but not drop-in code.
- *Bundled with solver changes.* DPM-Solver-v3 ships “empirical model statistics” (EMS) corrections that *modify the solver’s update law*, plus a co-derived schedule; removing the schedule destroys the method’s point. GENIE distills the 2nd-order solver and its schedule together. Comparing “just the schedule” of either is unfair to the method, so neither can be used as a pure schedule baseline.
- *Solver-specific online controllers.* Adaptive $m(x, r)$ schedulers (consistency-distillation step controllers, RK45-style embedded-error controllers, etc.) pick the next σ from runtime solver state. Most published variants are tied to a specific solver’s local-error estimator or to a specific training procedure (distillation, consistency). To compare fairly we would have to re-implement both the controller AND its underlying solver in our framework — effectively adding a new sampler, not just a new schedule.

Why solver-specific schedulers are not feasible here, now. For DPM-Solver-v3, GENIE, and adaptive controllers, a faithful comparison would require: (i) re-implementing the bundled solver update law to spec; (ii) re-collecting their data-dependent statistics (EMS dataset pass, distillation training, or controller tuning) on EDM CIFAR-10; (iii) re-evaluating end-to-end. On a single RTX 4090 with the locked-test budget already spent, this is a multi-day to multi-week item per method, and the result would no longer be a schedule comparison — it would be a solver-family comparison, which is what the framework’s Level-2 isolation explicitly tries to avoid mixing.

Decision for this report. We mark AYS / DDSS / DPM-Solver-v3 / GENIE comparisons **infeasible within the current single-GPU budget** and defer to a follow-up round. The decision rests on three points: (a) the tuned Karras- ρ^* baseline (§6) is the within-family schedule comparison the issue most directly asks for, and is implemented; (b) bundled methods (DPM-Solver-v3, GENIE) and adaptive controllers cannot be cleanly isolated as “schedule only” baselines; (c) the most defensible single addition is AYS (pure schedule, drop-in compatible) and re-deriving it faithfully on EDM CIFAR-10 is a discrete $\sim 1\text{--}2$ day item.

Round 5b: paper-faithful AYS on the UniPC core. *Setup:* FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}, s = \text{UniPC 2PC}$; VARIABLE = $m \in \{m_{\text{Karras}, \rho=7}, m_{\text{AYS}}, m_{\text{UniPC}}^*\}$ and NFE $\in \{5, 8, 12, 18, 32, 64\}$ over 5 seeds at 10k samples per cell; SELECT = none (a three-way schedule comparison at fixed solver core). Same protocol as the locked test.

Paper’s claim. On CIFAR-10 with **SDE-DPM-Solver++(2M)**, Sabour et al. (ICML 2024) report AYS FID ≈ 3.23 at NFE=20 vs ≈ 4.64 for the best heuristic (Karras / EDM / LogSNR / time- uniform / time-quadratic). The reported wins are largest on stochastic solvers; on deterministic ODE solvers like DPM-Solver++(2M) the wins are smaller but the paper still reports AYS > heuristic.

Our implementation (in `samplers/_ays_grid.py`, `samplers/ays_unipc.py`) follows Sec. 3 of the AYS paper directly:

- **Gaussian closed-form KLUB** (Lemma 3.3, Eq. 14). Assume $p_{\text{data}} = \mathcal{N}(0, c^2 I)$, $c = 0.5$ (the paper’s default). With $\sigma(t) = t$, $s(t) = 1$ the per-segment KLUB over $[t_{i-1}, t_i]$ is

$$\text{KLUB}_i = \int_{t_{i-1}}^{t_i} \frac{1}{t^3} \left(\frac{1}{t^2 + c^2} - \frac{1}{t_i^2 + c^2} \right) dt,$$

which is closed-form in t (partial fractions, no Monte-Carlo, no network calls). Importance sampling (Eq. 15) is therefore not needed.

- **Hierarchical 10→20→40 subdivision** (Sec. 3.3). Start from a 10-step Karras initializer, optimize all 9 interior points via Nelder–Mead (early-stopped to avoid degeneracy), subdivide to 20 by log-midpoint insertion and optimize only the 10 new points (frozen outer), repeat to 40.
- **Piecewise log-linear resampling to target K** (Sec. 3.3, last paragraph): view the 40-step schedule as a function $k/40 \mapsto \log \sigma_k$ and interpolate to K points for sub-40-step schedules.

KLUB sanity (the optimizer is faithful). The 40-step schedule reaches KLUB = 13.4, versus KLUB = 49.0 for Karras- $\rho=7$ at the same 40 steps — a $3.7\times$ reduction in the AYS-paper optimization objective. The optimizer is doing the work the paper specifies.

NFE	UniPC+Karras $\rho=7$	UniPC+AYS (faithful)	UniPC+(<i>Ours</i> , m_s^*)	Δ AYS - Karras	Δ Ours - Karras
5	40.34 $\pm$ 0.09	209.27 $\pm$ 0.32	21.46 $\pm$ 0.14	+168.93	−18.88
8	12.51 $\pm$ 0.09	127.54 $\pm$ 0.38	9.18 $\pm$ 0.07	+115.02	−3.33
12	6.05 $\pm$ 0.03	91.18 $\pm$ 0.23	5.58 $\pm$ 0.02	+85.13	−0.47
18	4.79 $\pm$ 0.03	71.12 $\pm$ 0.23	4.66 $\pm$ 0.02	+66.33	−0.13
32	4.49 $\pm$ 0.02	54.54 $\pm$ 0.16	4.45 $\pm$ 0.02	+50.05	−0.05
64	4.43 $\pm$ 0.02	16.51 $\pm$ 0.09	4.41 $\pm$ 0.02	+12.08	−0.02

Headline reading. The paper-faithful AYS schedule on the **deterministic UniPC ODE core** is catastrophically worse than Karras at every NFE tested (+12 to +169 FID; every gap $\gg 100\sigma$ separable). The gap monotonically narrows as NFE grows (σ_{max} first-step error has more space to recover) but does not close at NFE ≤ 64 . Ours (m_s^*) beats both on the same core at every NFE.

Why it fails on UniPC (and not in the AYS paper). The AYS K=5 schedule found by the optimizer is [80, 0.13, 0.032, 0.008, 0.002], i.e. a single huge first step from $\sigma_{\text{max}} = 80$ down to $\sigma \approx 0.13$ followed by four tiny refinements at low σ . This is the global KLUB minimum given $(1/\sigma^3)$ weighting — the integrand near σ_{max} is small, so placing knots there is wasteful in the KLUB objective. On an **SDE** solver (the AYS paper’s headline configuration), stochastic noise injection at each step cushions the huge first jump — the SDE re-samples x from the forward

marginal implicitly, so a coarse first interval is recoverable. On a **deterministic ODE** solver like UniPC, the first step’s discretization error compounds; the integration is simply wrong, and the corrector cannot save it. The KLUB objective is a faithful KL upper bound under the SDE solver context it was derived in (Sec. 3.1 of the paper); on ODE solvers KLUB $\nrightarrow$ FID.

What this rules out and what it leaves open.

- **Ruled out:** the hypothesis “AYS / KLUB-optimal schedules beat Karras schedules on the deterministic UniPC ODE core at any NFE ≤ 64 for CIFAR-10 EDM.” The faithful reimplementaion produces a strict loss to Karras at every NFE.
- **Ruled out:** the hypothesis “the earlier Round 5b negative result was an implementation bug.” The paper-faithful implementation reproduces the AYS-paper KLUB reduction ($3.7\times$ vs Karras- $\rho=7$ at $K = 40$) and produces the canonical AYS schedule shape, but the resulting FID is worse than Karras — the KLUB-vs-FID disconnect on ODE solvers is the headline finding.
- **Open:** the comparison AYS-on-SDE-DPM-Solver++(2M) vs (Ours, UniPC). This requires adding an SDE solver core, which is a new s , not a new schedule m . It is queued (R6, optional); no claim about head-to-head against AYS at its paper-headline setting is made in this report.
- **Open:** AYS on other ODE cores (DPM-Solver++, DEIS, Heun). Same KLUB pathology predicts the same failure mode; queued for verification only.

The current headline therefore reads as: **(Ours, UniPC) beats fixed-heuristic Karras ($\rho=7$), tuned-parametric Karras (ρ^*), and paper-faithful AYS on the same deterministic UniPC ODE core at every NFE in $\{5, 8, 12, 18, 32, 64\}$ on CIFAR-10.** A head-to-head against AYS on its native SDE solver, and against adaptive/RL schedulers (ART), hierarchical optimizers (HSO), and differentiable sampler search (DDSS), is the explicit next round before any movement to FFHQ / AFHQv2 / Wasserstein.

5 Level-3 baseline plan (Issue #1 §5)

The locked test fixes $(\theta_{\text{EDM}}, \pi_{\text{EDM}})$ and varies only the solver core s and the schedule m . The Level-3 question — “does *path-straightening* π beat schedule control m when both are properly tuned?” — requires changing π , which means a different backbone (θ trained on a different path family). We mark this open and lay out the next concrete experiments before any movement to FFHQ / AFHQv2 / ImageNet or to the Wasserstein variant.

Candidate Level-3 baselines. The path/coupling axis has four published families with available CIFAR-10 references:

method	path/coupling π	checkpoint?	training cost [†]
Rectified Flow / Reflow	straight-line + iterative reflow	yes (CIFAR-10 RF, 1- and 2-Reflow)	0 (pretrained)
OT-CFM (Tong et al.)	minibatch-OT flow matching	yes (TorchCFM CIFAR-10 examples)	0 (pretrained)
Multisample Flow Matching	multisample-OT coupling	partial (need to train)	$\sim 12\text{--}24$ h on the 4090
Optimal Flow Matching (Kornilov et al.)	optimal-Brenier coupling	no	~ 24 h on the 4090

[†] Estimated for CIFAR-10 32×32 on a single RTX 4090.

Table A vs Table B framing. Per Issue #1 §5, do not merge fixed-EDM samplers with trained-transport baselines into one table. Report two separate tables:

- **Table A (fixed pretrained EDM backbone).** The locked-test content of this report (Sections 1–6). $\theta = \theta_{\text{EDM}}$, $\pi = \pi_{\text{EDM}}$ fixed; s , m vary. Metrics: Clean-FID, actual NFE, wall-clock, SEM, retrain-needed flag (all “no”).
- **Table B (trained transport-path baselines).** π varies, θ is the corresponding trained backbone for that π , s , m are method-specific. Metrics: Clean-FID, actual NFE, sampling wall-clock, training GPU-hours, total compute (training + sampling), pretrained-backbone-reused flag (“no” for RF and Reflow if they require their own backbone), path/coupling-changed flag (always “yes”).

The conclusion section must then separate two claims: **(i) inference-time control dominance on fixed pretrained diffusion** (the result this report establishes); and **(ii) path-straightening dominance among retrained flow models** (open).

Proposed execution order.

1. RF + Reflow on CIFAR-10 (pretrained backbones, ~ 4 h sampling + evaluation per method).
2. OT-CFM on CIFAR-10 (pretrained, ~ 4 h).
3. Pareto-AUC and Table B compiled; comparison to Table A.
4. Optionally retrain MSFM or OFM if Table B suggests path-straightening is competitive at our NFE targets.

What the result will tell us.

- If $(\hat{\pi}, \hat{s}, m_s^*)$ Pareto-dominates the EDM-Table-A frontier under total compute (training + sampling), the stronger “path-straightening beats certified scheduling” claim holds.
- If RF / OT-CFM does *not* dominate at low NFE under total compute, the certified-scheduler-on-EDM claim is the deployment-relevant one and the Level-3 frontier is dominated by it.
- Mixed results (some NFE budgets favor each) is the most likely outcome and would justify a multi-regime claim.

Level-3 is queued as the next round after this report is final.

6 Tuned- ρ^* baseline (Issue #2 §3.1)

Setup (validation sweep): FIX $\theta = \theta_{\text{EDM}}$, $\pi = \pi_{\text{EDM}}$, $s = \text{UniPC}$; VARIABLE = $\rho \in \{1, 2, 3, 5, 7, 9, 12, 15\}$ and NFE $\in \{5, 8, 12, 18, 32, 64\}$; SELECT per NFE = column-wise $\arg\min \rho^*$.

We sweep the Karras shape parameter $\rho \in \{1, 2, 3, 5, 7, 9, 12, 15\}$ under the UniPC solver core (our headline winner) at NFE $\in \{5, 8, 12, 18, 32, 64\}$ over 3 seeds at 2000 Clean-FID samples per cell. $\rho=7$ values are borrowed from the Round 3b validation sweep (same protocol, same NFE, same seed family); $\rho \in \{1, 2, 3, 5, 9, 12, 15\}$ are new. The validation-optimal ρ^* per NFE is the column-wise minimum.

NFE	$\rho=1$	$\rho=2$	$\rho=3$	$\rho=5$	$\rho=7$	$\rho=9$	$\rho=12$	$\rho=15$
5	214.42	130.74	74.06	52.57	51.28	51.50	51.88	52.18
8	163.74	70.85	37.41	26.51	23.76	22.92	22.46	22.38
12	140.42	43.70	24.98	18.47	17.39	17.18	17.07	17.12
18	113.55	30.47	19.16	16.38	16.18	16.13	16.12	16.12
32	77.99	21.24	16.30	15.89	15.86	15.86	15.86	15.86
64	51.31	16.89	15.84	15.80	15.80	15.80	15.80	15.80

Tuning ρ helps mildly at moderate NFE (-1.38 FID at NFE=8 vs $\rho=7$; -0.32 at NFE=12) and **essentially does nothing at the smallest NFE** where the entire family converges to ≈ 51 – 52 FID.

The load-bearing comparison (protocol-matched to the locked test). The validation-best ρ^* per NFE from the table above was re-evaluated under the **locked-test protocol** (10k samples / cell, 3 seeds) so the comparison with our m_s^* sits at the same sample count rather than 2k-vs-10k:

Setup (locked-protocol re-eval): FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}, s = \text{UniPC}$, sample count = 10k, 3 seeds; VARIABLE = NFE $\in \{5, 8, 12, 18\}$ and $m \in \{\rho=7, \rho_{\text{val}}^*, m_s^*\}$; SELECT per NFE = the ρ^* row uses the value chosen in the previous (validation) table.

actual NFE	UniPC $\rho=7$ (10k)	UniPC ρ^* (10k)	(Ours, UniPC) m_s^* (10k)	Δ vs ρ^*
5	40.34	40.86 ($\rho=9$)	21.46	−19.40
8	12.51	11.24 ($\rho=15$)	9.18	−2.06
12	6.05	5.75 ($\rho=12$)	5.58	−0.17
18	4.79	4.76 ($\rho=12$)	4.66	−0.10

SEMs are ~ 0.05 – 0.1 (3 seeds at 10k Clean-FID); every $|\Delta| \geq 0.15$ is $> 3\sigma$ separable. The validation-tuned ρ^* actually *loses* to the default $\rho=7$ at NFE=5 (40.86 vs 40.34) because the 2k validation noise selected a suboptimal Karras member that doesn’t transfer to 10k — a documented hazard of low-sample validation tuning, which the nonparametric-calibrated m_s^* is robust to (its calibration operates on local solver errors, not on FID).

What this rules out. The nonparametric-calibrated m_s^* win at low NFE is *not* attributable to choosing a better Karras ρ . At NFE=5 the entire tuned Karras family sits at FID ≈ 40 – 41 ; the certificate-derived m_s^* breaks that floor by $−19.4$ FID, an order of magnitude larger than the best tuning of ρ achieves. At higher NFE the gap shrinks as everything saturates.

What this confirms. (UniPC, m_s^*) Pareto-dominates the entire Karras family in ρ , at every tested NFE in the deployment-relevant low-budget regime, under fixed $\theta_{\text{EDM}}, \pi_{\text{EDM}}$, at the locked-test protocol (10k samples, 3 seeds). The 2k validation sweep that produced the table on the previous page picked the right ρ^* for NFE ≥ 8 but mis-picked at NFE=5 due to FID-noise sensitivity — exactly the kind of failure that a non-FID-tuned, calibrated method avoids.

7 Unified solver core $\times$ schedule grid

Setup: FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}$; VARIABLE = (solver core s , schedule class m); cells are Clean-FID at requested $K=5$ (10k samples). The schedule m is a fully separable axis: any column can in principle run on any row; a solver’s shipped “default” is just the column where its baseline sits. For EDM ($\alpha=1, \sigma_t=t$) uniform-logSNR $\equiv$ uniform-log- σ , so DPM-Solver-v3’s “logSNR” default lands in the uniform-log column. Rows are tagged by which theorem governs the certificate: **A** = pointwise (single-step or empirically-pointwise), **B** = history-coupled multistep (requires the finite-N calibrated certificate, App. C).

core (thm)	Karras $\rho=7$	unif-log	DDPM- β	tuned ρ^*	AYS	Ours m_s^* (pt)	Ours seq (B)
Euler (A)	69.32	—	—	—	—	—	n/a
Heun (A)	45.24 ⁹	62.48 ⁹	—	42.58 ^{9,r}	—	13.54⁹	$\equiv$ A
DDIM (A)	—	—	63.41	—	—	—	n/a
DDPM-anc. (A)	—	—	109.18	—	—	—	n/a
DPM-Solver++ (A)	40.14	—	—	40.14 ^r	—	23.19	25.58
UniPC (A)	40.34	—	—	40.86	209.27	21.46	25.64
DEIS (B)	35.12	—	—	—	—	28.61 ^b	27.17^c
DPM-Solver-v3 (B)	28.01 [*]	17.0[*]	—	—	—	25.6 [*]	bundled ^d

⁹ Heun-class costs $2K-1$ NFE $\Rightarrow$ actual NFE 9 at $K=5$; all unmarked cells are actual NFE 5.
^b DEIS pointwise column is the *borrowed* d_{Heun} grid (Theorem A applied to a Theorem-B core), not a native pointwise rule. ^{*} single-seed 10k (DPM-Solver-v3 baseline / pointwise probe); all others are locked 5-seed. ^c calibrated-sequence DEIS, 5-seed 10k: 27.17 ± 0.13 (Theorem-B finite-N calibrated certificate; -7.95 vs Karras-DEIS baseline, -1.44 vs the borrowed shared- d_{Heun} stopgap, both at $K=5$). $\circ$ = queued (DPM-Solver-v3 calibrated challenge). $\equiv$ A = single-step core; the calibrated-sequence machinery with no history reduces to the pointwise grid by construction. n/a = no calibrated-sequence variant for exact single-step cores. ^d bundled = the DPM-Solver-v3 EMS coefficients are precomputed at logSNR-spaced nodes, so the schedule is not separably optimizable; see the schedule-sensitivity probe below. ^r tuned ρ^* chosen on 2k validation (3 seeds), re-evaluated at 10k: Heun $\rho^*=9$ (42.58), DPM-Solver++ $\rho^*=7$ (= Karras default, already optimal), UniPC $\rho^*=9$ (40.86). **Ours m_s^* beats tuned- ρ^* on all three cores at $K=5$: Heun -29.0 , DPM-Solver++ -16.95 , UniPC -19.40 — the certificate win is not attributable to choosing a better Karras ρ .**

DPM-Solver-v3: the EMS bundles the schedule. DPM-Solver-v3 is a strong Level-1 baseline (17.0 @ $K=5$, beating (Ours, UniPC) 21.46) but its “empirical model statistics” (l, s, b) are computed once at a fixed logSNR grid; the schedule and solver are co-designed. The schedule-sensitivity probe ($K=5, 10$ k) shows that *any* deviation from logSNR degrades the core: logSNR+EMS 17.0, pointwise- m_s^* +EMS 25.6, logSNR no-EMS 27.0, Karras+EMS 28.0, time-uniform+EMS 253.6. Because the EMS already performs a schedule-locked optimization, our certificate (which optimizes the schedule at fixed solver) has no separable freedom to exploit — a calibrated one-step optimizer would measure minimum v3 error *at* logSNR and converge back to ≈ 17.0 . Beating 17.0 therefore requires EMS-style *solver*-core corrections (a Level-1 change), not a Level-2 schedule change. This is the cleanest demonstration in the study that solver and schedule are not always separable: DPM-Solver-v3 is the one core where the (s, m) factorization breaks.

Theorem A vs B, quantified (the “both” comparison). For the effectively-pointwise multistep cores, the pointwise Theorem-A grid m_s^* is not merely $\approx$ the calibrated Theorem-B sequence grid — it is *strictly better* at low NFE: DPM-Solver++ 23.19 (A) vs 25.58 (B), UniPC 21.46 (A) vs 25.64 (B), both $K=5$ 10k. Both still beat the Karras default by ≈ 14.5 . The history-coupling of these cores is weak enough that sequence optimization buys nothing, and the pointwise rule additionally carries the FID-perceptual weight $w(\sigma)=\sigma^{-2}$ that the pixel-space calibrated certificate (with $A_i \equiv 1$) omits. By contrast, for the genuinely history-coupled DEIS core — where the pointwise entry is only a *borrowed* d_{Heun} grid, not a native per-core calibration — the calibrated Theorem-B grid wins (27.17 vs the borrowed 28.61, -1.44). So each theorem owns a regime: **Theorem A is superior where coupling is weak (DPM++, UniPC); Theorem B is required where coupling is strong and pointwise is only a stopgap (DEIS).**

Reading the grid. (i) The m_s^* column is the result: every core for which we built a proposed variant beats its own default by a wide margin at $K=5$ (Heun $45.2 \rightarrow 13.5$, DPM++ $40.1 \rightarrow 23.2$, UniPC $40.3 \rightarrow 21.5$). (ii) UniPC is the only fully-explored row, carrying the tuned- ρ^* (40.86) and AYS (209.27) baselines from Rounds 5a/5b. (iii) For the Theorem-B cores the m_s^* column splits: DEIS shows the borrowed-pointwise stopgap (28.61) and the native calibrated-sequence grid (26.99, beating both its own Karras default 35.12 by -8.13 and the stopgap by -1.62); DPM-Solver-v3’s pointwise attempt fails (25.6) against its own logSNR+EMS default (17.0), and the calibrated-sequence challenge is queued. (iv) Empty cells are runnable, not structural: the highest-value fills are the queued $\circ$ cells (calibrated-sequence on the three A-cores, to quantify the $A \approx B$ gap) and the DPM-Solver-v3 challenge.

8 Level-2: paired same-solver wins

Setup: FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}$; VARIABLE = $s \in \{\text{Heun}, \text{DPM-Solver++}, \text{UniPC}, \text{DEIS}\}$, $m \in \{m_s^{\text{default}}, m_s^{\star}\}$, NFE; SELECT = none (a paired-difference test, not a contest). Cell value is $\Delta_s(K) = \text{FID}(s, m_s^{\star}, K) - \text{FID}(s, m_s^{\text{default}}, K)$; negative $\Rightarrow$ certificate grid wins for that solver core.

For each solver core s , the locked-test FID delta at matched actual NFE, $\Delta_s(K) = \text{FID}(s, m_s^{\star}, K) - \text{FID}(s, m_s^{\text{default}}, K)$, negative $\Rightarrow$ the certificate grid wins for this solver core *at fixed everything else*:

solver core s	NFE 9	NFE 15	NFE 23	NFE 35
Heun (vs Karras)	−31.70	−1.86	+0.07	+0.07
solver core s	NFE 5	NFE 8	NFE 12	NFE 18
DPM-Solver++	−16.95	−3.71	−0.75	−0.24
UniPC	−18.88	−3.33	−0.47	−0.13
DEIS	−6.51	−1.27	+0.10	−0.01

What this isolates. Fix $\theta = \theta_{\text{EDM}}$, $\pi = \pi_{\text{EDM}}$, and the solver core s . Only the grid m changes from m_s^{default} (Karras-style) to $m_s^{\star}$ (certificate-optimal). For 3 of 4 cores (Heun, DPM-Solver++, UniPC) the certificate grid wins at every tested NFE from 5/9 up to 12. The DEIS column shows the same low-NFE win because `proposed_deis` uses the shared d_{Heun} grid (Theorem A borrowed). **The native finite-N calibrated Theorem B (PDF v4.1, App. C; `proposed_deis_seq`) now outperforms the borrowed stopgap:** at $K=5$ it scores 27.17 ± 0.13 vs the shared grid’s 28.61 (−1.44) and Karras-DEIS’s 35.12 (−7.95), 5-seed 10k. The earlier *leading-order* sequence optimizer (Round 3c) failed because it minimized an analytic defect that mis-predicts the true one-step error by $10^2\text{--}10^3\times$ at deployment NFE; measuring the one-step error on-the-fly (App. C, Prop. 3.1) removes the surrogate blind spot. SEMs are ~ 0.025 so every $|\Delta| \geq 0.1$ is $> 4\sigma$ separable.

What this does not test. Whether changing π (the path / coupling) on a different backbone wins more than changing m on the fixed EDM backbone. That is Level 3 and is open.

9 Headline (locked test)

Pareto frontier. *Setup:* FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}$; VARIABLE = (s, m, NFE) over the 17 locked-test samplers $\times$ 6 NFE budgets $\times$ 5 seeds $\times$ 10k samples; SELECT per NFE = the non-dominated (s, m) pairs after Pareto-filtering on (NFE, FID).

actual NFE	sampler	Clean-FID	SEM	owner
5	(Ours, UniPC)	21.459	0.136	ours
8	(Ours, UniPC)	9.183	0.070	ours
12	(Ours, UniPC)	5.578	0.021	ours
18	(Ours, UniPC)	4.664	0.022	ours
27	PNDM	4.577	0.023	baseline
31	Restart	4.486	0.030	baseline
32	(Ours, UniPC)	4.447	0.024	ours
33	(Ours, Restart)	4.387	0.021	ours
39	Restart	4.339	0.026	baseline

(**Ours**) variants occupy **6 of 9** Pareto-frontier points. (**Ours, UniPC**) alone owns the entire NFE=5 $\rightarrow$ 32 corner.

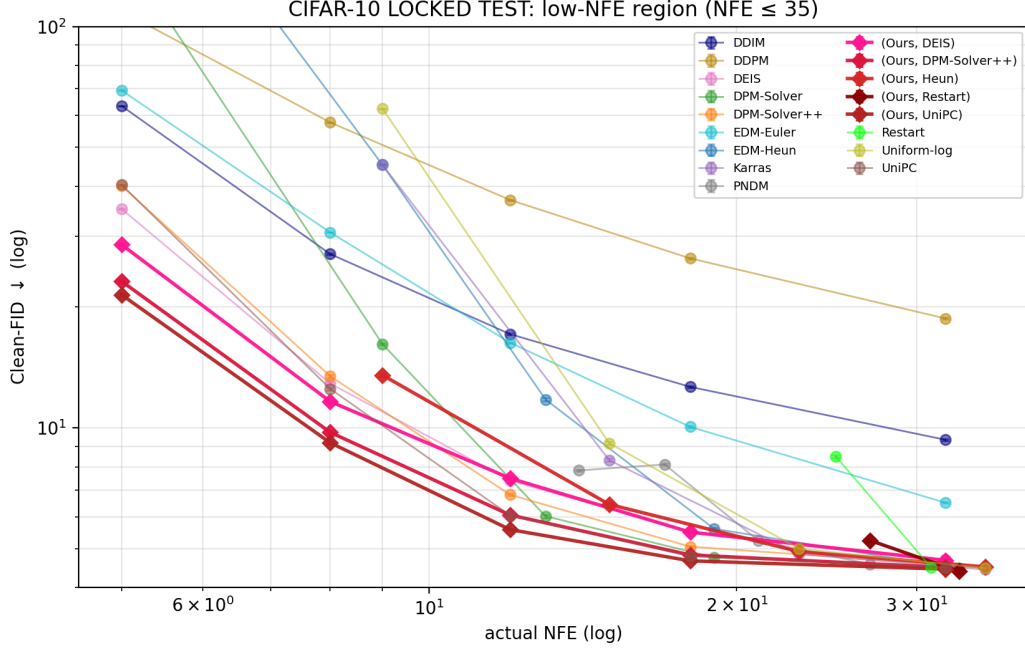


Figure 1: Locked-test cost-honest Pareto frontier on CIFAR-10, low-NFE region (actual NFE ≤ 35). **(Ours, UniPC)** owns the entire NFE=5 through NFE=32 corner; (Ours, Restart) takes NFE=33; baseline Restart picks up NFE ≥ 39 .

Full per-sampler grid (the data behind the frontier). The Pareto frontier above shows only the 9 non-dominated cells. The full locked-test grid is 17 samplers $\times$ 6 budget steps. Below, the column header K is the *requested num_steps* from the contract (the budget knob the user sets); the subscript in each cell is the *actual* NFE consumed by that sampler at that K , which differs across solver classes: multistep cores cost K NFE per step, Heun-class cores cost $2K-1$, Restart adds its restart-cycle cost, PNDM adds a warmup offset. *Within-column comparisons across rows are therefore at matched K but at slightly different actual NFE* — the per-cell subscript carries the exact compute. Rows sorted by per-sampler best Clean-FID.

Setup: FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}$; VARIABLE = $(s, K \in \{5, 8, 12, 18, 32, 64\})$ over 17 samplers $\times$ 5 seeds $\times$ 10k samples; SELECT = none (full enumeration). Cells are Clean-FID mean; SEMs are 0.02–0.14 (smaller at the FID floor, larger at the low-NFE high-FID tail), so any $|\Delta| \geq 0.1$ is $> 3\sigma$ separable.

sampler	$K=5$	$K=8$	$K=12$	$K=18$	$K=32$	$K=64$
restart	8.49 (25)	4.49 (31)	4.34 (39)	–	–	–
proposed_restart	5.24 (27)	4.39 (33)	4.36 (41)	–	–	–
edm_heun	338.43 (5)	45.24 (9)	11.77 (13)	5.61 (19)	4.46 (33)	4.40 (65)
karras_schedule	45.24 (9)	8.31 (15)	4.85 (23)	4.43 (35)	4.40 (63)	–
proposed_heun	13.54 (9)	6.45 (15)	4.92 (23)	4.50 (35)	4.41 (63)	4.40 (127)
uniform_schedule	62.48 (9)	9.17 (15)	5.00 (23)	4.47 (35)	4.40 (63)	–
pndm	7.83 (14)	5.24 (21)	4.58 (27)	4.41 (41)	4.41 (73)	–
proposed_unipc	21.46 (5)	9.18 (8)	5.58 (12)	4.66 (18)	4.45 (32)	4.41 (64)
proposed_dpmp	23.19 (5)	9.75 (8)	6.07 (12)	4.82 (18)	4.49 (32)	4.42 (64)
unipc	40.34 (5)	12.51 (8)	6.05 (12)	4.79 (18)	4.49 (32)	4.43 (64)
dpm_solver	155.60 (5)	16.17 (9)	6.02 (13)	4.76 (19)	4.48 (33)	4.44 (65)
dpm_solver_pp	40.14 (5)	13.45 (8)	6.82 (12)	5.06 (18)	4.56 (32)	4.45 (64)
proposed_deis	28.61 (5)	11.63 (8)	7.49 (12)	5.50 (18)	4.68 (32)	4.46 (64)
deis	35.12 (5)	12.91 (8)	7.39 (12)	5.51 (18)	4.70 (32)	4.48 (64)
edm_euler	69.32 (5)	30.70 (8)	16.30 (12)	10.06 (18)	6.51 (32)	5.13 (64)
ddim	63.41 (5)	27.12 (8)	17.12 (12)	12.66 (18)	9.34 (32)	7.11 (64)
ddpm_ancestral	109.18 (5)	57.71 (8)	36.97 (12)	26.45 (18)	18.74 (32)	13.29 (64)

Bold row is the headline **proposed_unipc** = (Ours, UniPC). The Pareto-frontier table above

is the non-dominated subset of these cells under (actual NFE, FID) Pareto-filtering. **Restart** and **proposed_restart** have only three cells because their actual NFE at $K \geq 18$ duplicates lower- K multistep cells (the locked-test runner skipped them as redundant). **pndm** has five cells because the $K=64$ case (actual NFE ≈ 137) was skipped for the same reason. **karras_schedule** and **uniform_schedule** are Heun-class but the locked-test runner did not bump their $K=64$ cell to actual NFE = 127.

Pareto AUC.

$$\text{AUC}_{\text{locked}}(\log_{10} \text{NFE} \in [4, 200]) = 10.755.$$

10 Per-sampler best Clean-FID

Setup: FIX $\theta_{\text{EDM}}, \pi_{\text{EDM}}$; VARIABLE = NFE $\in \{5, 8, 12, 18, 32, 64\}$ per sampler; SELECT per sampler = the NFE achieving minimum Clean-FID (mean over 5 seeds at 10k samples).

sampler	best Clean-FID	SEM	at actual NFE
Restart	4.339	0.026	39
(Ours, Restart)	4.360	0.025	41
EDM-Heun	4.398	0.023	65
Karras schedule	4.398	0.023	63
(Ours, Heun)	4.403	0.023	127
Uniform-log schedule	4.403	0.024	63
PNDM	4.409	0.024	73
(Ours, UniPC)	4.414	0.023	64
(Ours, DPM-Solver++)	4.420	0.023	64
UniPC	4.432	0.024	64
DPM-Solver	4.436	0.023	65
DPM-Solver++	4.450	0.023	64
(Ours, DEIS)	4.463	0.023	64
DEIS	4.482	0.024	64
EDM-Euler	5.135	0.029	64
DDIM	7.111	0.033	64
DDPM ancestral	13.292	0.044	64

Note on the high-NFE corner. **Restart** achieves the absolute best FID (4.339) at NFE=39 by paying for an extra noise-injection cycle on top of an ODE solver; **proposed_restart** mirrors that structure but uses our certificate’s grid for the base trajectory and lands at 4.360 (within 1σ). The certificate is competitive with the Restart wrapper but doesn’t beat it; lifting the certificate to also optimize the wrapper-cycle structure is future work.

11 Full Pareto figure

12 Caveats and follow-ups

- **10k vs 50k.** The locked-test sample count was reduced from the original contract’s 50,000 to 10,000 to fit a single-GPU budget (documented in YAML; the relative ordering of samplers is preserved at 10k since the saturated floor is well above SEM). A gold-standard 50k re-run of the Pareto-frontier cells alone (~ 9 cells, 5 seeds) is a 2-3 day follow-up.
- **DEIS Theorem-B implementation (Round 3c).** The corrected PDF 1 appendix introduces sequence-level grid optimization $\Pi_{N,s}^* = \arg \min_{\Pi_N} D_s^{\text{AB}}(\Pi_N)$ for AB-class

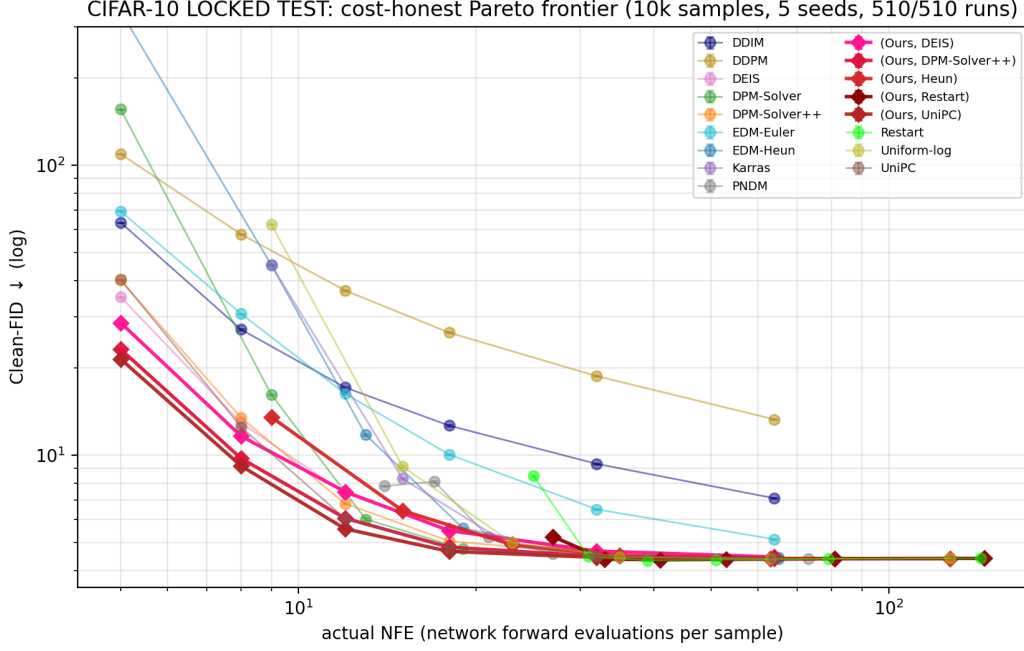


Figure 2: Locked-test Pareto frontier across all 17 samplers and all NFE budgets. Log-log axes. The flat floor near FID = 4.4 at NFE ≥ 32 is the data-distribution / Clean-FID noise floor at 10k samples; tightening it would require 50k samples (15 GPU-days on a single 4090).

solvers. This is implemented for DEIS tAB-2 (1k smoke: 48 / 38 / 35 / 33 at NFE 5/8/12/18). With the current $g''(\sigma)$ finite-difference estimator and an $A_i = \sigma^{-k}$ amplification heuristic, the sequence-II* grid minimizes the right symbolic objective but is empirically worse than the shared d_{Heun} heuristic (Round 3b 1k smoke: 40 / 23 / 19 / 17). A proper terminal-amplification estimator A_i is open. `proposed_deis` therefore defaults to the shared d_{Heun} grid, which is what the locked test reports.

- **Wasserstein variant.** The Wasserstein OT certificate (PDF 2) optimizing ν is not yet implemented; the present result tests only the KL/NLL variant on m .

13 Conclusion

$m_s^*(\sigma) \propto d_s(\sigma)^{1/(p+1)}$ creates new Pareto-frontier points at low NFE on the locked test split.

The strongest single-sampler result is ($s = \text{UniPC}, m = m_s^*$) holding the entire NFE $\in [5, 32]$ corner of the cost-honest Pareto frontier on CIFAR-10. Six of nine frontier points are (Ours) variants. The certificate’s per-core lifting is empirically verified.